

scheduled-work-engine

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A decoupled, project-agnostic **scheduled-work** / **reminder tracking service**. It records background-queue work items and lets an autonomous agent — when a REMINDER fires — re-verify **uncertain** / **blocked** / **overdue** work before reporting it done (constitution §11.4.6 no-guessing, §11.4.108 verify-before-report). It is the backing infrastructure for the REMINDER / BACKGROUND action-prefix (§11.4.140).

Surfaces (one implementation, multiple transports)

Surface	Transport	Entry
REST	HTTP/3 (QUIC) + h2/h1 fallback, brotli	serve → /api/v1/...
MCP	HTTP/3 + h2/h1 (JSON-RPC 2.0)	serve → POST /mcp
MCP	stdio (Claude Code plugin)	mcp-stdio
ACP-equivalent	stdio (JSON-RPC 2.0)	acp-stdio

Router = **gin** (gin.Engine is served identically over h1/h2/h3); HTTP/3 = **github.com/quic-go/quic-go/http3**; compression = **github.com/andybalholm/brotli** middleware.

Build

```
go build -o bin/scheduled-work ./cmd/scheduled-work # or:
bash ../../plugins/scheduled-work/build.sh
```

The binary is a build derivative (gitignored per §11.4.30; regen §11.4.77).

Run

```
# REST + MCP-over-HTTP on HTTP/3 (+ h2/h1) with a self-signed dev
cert:
SWQ_ADDR=127.0.0.1:8787 ./bin/scheduled-work serve

# MCP over stdio (for a Claude Code plugin / any MCP stdio
client):
./bin/scheduled-work mcp-stdio

# ACP-equivalent JSON-RPC over stdio:
./bin/scheduled-work acp-stdio
```

Configuration (env — §11.4.28 decoupling: no hardcoded consumer data)

Var	Default	Meaning
SWQ_ADDR	127.0.0.1:8787	listen address (TCP for h1/h2, UDP for h3)
SWQ_DB	./scheduled_work.json	JSON store path
SWQ_TLS_CERT / SWQ_TLS_KEY	(self-signed dev)	production TLS pair
SWQ_BROTLI	5	brotnli quality 0–11, or -1 to disable
SWQ_H3	1	1 enable HTTP/3, 0 = h1/h2 only

REST API

Method	Path	Purpose
GET	/healthz	liveness + item count
POST	/api/v1/items	create {title, description?, status?, due_at?, tags?, source?, notes?}
GET	/api/v1/items?status=	list (effective-status filter, incl. overdue)

Method	Path	Purpose
GET	/api/v1/items/ overdue	open items past due
GET	/api/v1/items/needs- verification	open blocked/ uncertain/overdue
GET	/api/v1/items/:id	one item
PATCH	/api/v1/items/:id/ status	{status, notes?}
POST	/api/v1/items/:id/ done	{notes?} — mark done (put evidence in notes)
DELETE	/api/v1/items/:id	delete

MCP tools

create_work_item, list_work_items, get_work_item,
update_work_item_status, mark_work_item_done, list_overdue_work,
list_needs_verification. See skills/scheduled-work-queue/skill.md.

Status model (closed set)

scheduled · in-progress · blocked · uncertain · done · cancelled, plus a **derived** overdue (an open item past due_at, computed on read, never stored — §11.4.6). blocked / uncertain / overdue are the states a REMINDER MUST confirm before reporting done.

Testing

```
go test ./... # unit (store, mcp)
go test -race ./test/ # integration: REAL HTTP/3 +
    brotli round-trip
bash test/meta_mutation.sh # §1.1 paired mutation (RED-
    on-broken / GREEN-on-restore)
bash test/live_smoke.sh
    # live captured-evidence smoke (REST TLS + MCP stdio)
```

See DESIGN.md for the store-choice justification, the ACP honesty note, and the proposed governance-anchor handoff.